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United States Patent	12415673
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Pearce; Brooks

Vapor and liquid recovery tower

Abstract

A vapor and liquid recovery tower (VLRT) includes a top section adapted to receive a liquid stream with entrained gas that gravity flows down through the VLRT, and a packing section comprising packing material adapted to release the entrained gas from the liquid stream to produce a VLRT liquid output and a VLRT gas output. The VLRT may further include a lower section adapted to receive a gas stream that rises up through the packing section, wherein the packing material is further adapted to increase surface area contact between the liquid stream and the gas stream, and wherein the VLRT gas output includes the entrained gas and the gas stream.

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Appl. No.:	18/119102
Filed:	March 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230286737 A1	Sep. 14, 2023

Related U.S. Application Data

continuation-in-part parent-doc WO PCT/US2022/030744 20220524 PENDING child-doc US 18119102
continuation-in-part parent-doc US 17488819 20210929 US 11725152 child-doc US PCT/US2022/030744

Publication Classification

Int. Cl.: **B65D90/30** (20060101); **C10G33/06** (20060101)

U.S. Cl.:

CPC **B65D90/30** (20130101); **C10G33/06** (20130101);

Field of Classification Search

CPC: B65D (90/30); C10G (33/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2780304	12/1956	Pew	208/187	C10G 33/06
3505194	12/1969	Martin	204/570	C10G 33/06
2010/0307834	12/2009	Largent	N/A	N/A
2016/0008742	12/2015	Adler et al.	N/A	N/A
2017/0121610	12/2016	Niccum et al.	N/A	N/A
2017/0369791	12/2016	Khan	N/A	C10G 31/08
2020/0165528	12/2019	Nanda et al.	N/A	N/A

OTHER PUBLICATIONS

PCT International Search Report for related PCT Application No. PCT/US2024/018986, dated Jul. 8, 2024, 4 pages. cited by applicant
PCT Written Opinion of the International Searching Authority for related PCT Application No. PCT/US2024/018986, dated Jul. 8, 2024, 5 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of PCT International Application No. PCT/US2022/030744 entitled “System and Method of Reducing Emissions and Increasing Swell in an Oil Conditioning Process” and filed on May 24, 2022, which claims the benefit of priority to: U.S. patent application Ser. No. 17/488,819 entitled “System and Method of Reducing Emissions and Increasing Swell in an Oil Conditioning Process” and filed on Sep. 29, 2021, U.S. Provisional Patent Application Ser. No. 63/196,154 entitled “System and Method of Reducing Emissions and Increasing Swell in an Oil Conditioning Process”

BACKGROUND

- (1) The present invention relates to oil and gas production, and more particularly to technology for conditioning or stabilization of live crude oils at the outlet of the extraction well.
- (2) The output of oil and gas well-heads typically includes oil, water, and gas, often in an emulsion, at pressures between approximately 150 PSI and 1,500 PSI (10 and 100 bars). The process partial distillation of live crude oil and reducing the well-head pressure according to API standards is referred to as stabilization.
- (3) In a typical stabilization process, illustrated in FIG. 1A, a live crude oil stream **1320** (including oil, gas, and water) from a wellhead **1220** first goes to a separator **1230**. The separator **1230** reduces the pressure of the live crude oil stream **1320** and outputs an oil stream **1332**, a gas stream **1334**, and a water stream **1336**. Among the output streams from the separator **1230**, gas **1334** released from emulsion can go directly to sale, water **1336** removed from the bottom can go to a storage and/or treatment facility, and oil stream **1332** can go to a tank **1235** for holding for additional stabilization, as oil stream **1332** typically contains light hydrocarbons and water and is at higher than atmospheric pressure, after processing only by the separator **1230**.
- (4) In many conventional systems (FIG. 1A), oil stream **1332** is pumped from the tank **1235** into a heater treater **1240**, which typically outputs an oil stream **1342**, a gas stream **1334**, and a water stream (not shown in FIG. 1A). Gas stream **1334** can go to a vapor recovery compressor **1260** or like device, as gas is moved for sale. Oil stream **1342** can go to a stabilizer tower **1250**, which can output a gas stream **1354** and an oil stream **1352**. Gas stream **1354** can go to a vapor recovery compressor **1260** or like device, as gas stream **1354** is moved for sale. Oil stream **1352** is stabilized to the degree that it can be stored in a conventional stabilized crude oil tank **1265** at or near atmospheric pressure. Each of the prior art components are explained below.
- (5) In general, a separator is a pressure vessel that, in a two-phase unit, receives a process flow for a retention time that allows lighter hydrocarbons to escape from the flow stream as a gas. In a three-phase separator, water also settles out from the oil for removal beneath the oil outlet of the separator. A separator generally includes internal portions or devices to promote separation, sometimes referred to as gravity settling, of the oil and water and release the gas. Often a mist extractor is used to remove liquid droplets from the gas. A separator often includes a liquid-level controller to help control internal fluid levels and a means, such as a back-pressure valve, to control internal pressure.
- (6) Often several stages of separation are employed, depending on the particular process variables of the site, to reduce pressure in stages. The separator is sometimes referred to as a Trap, a Knockout vessel, a flash chamber, an expansion vessel, or the like. Typically, the separator **1230** is near wellhead **1220**, although in some installations may be located a mile away. Many separator designs have been developed, and the explanation of separator in general and/or separator **1230** is not intended to be limiting in any way.
- (7) In general, a heater treater, such as heater treater **1240**, is a 3-phase vessel that utilizes heat and mechanical separation devices for further separating the oil stream **1332** from the separator **1230** into an oil stream **1342**, a gas stream **1334**, and a water stream (not shown in FIG. 1A). Heater treaters typically include a degassing section, a heating section, differential oil control, and a coalescing section, although not every section is required to meet the definition of a heater treater.
- (8) Oil stream **1332** (or untreated, live oil in installations that do not have an initial separator, such as separator **1230**) enters the degassing section via an inlet—often at the top of the vessel. Gases that are easily released are vented into a gas collection line that often includes a mist extractor, to produce gas stream **1334**. Water within the oil stream **1332** can drop to the bottom of the vessel for removal at a water outlet. After initial degassing, the emulsion passes into a heating section, which

often includes a tube-type heat exchanger heated to a temperature of approximately 100 to 160 degrees F. Some heater treaters have a section containing a filtering medium to screen solid particles out of the oil. This process of heating the crude at this stage decreases the oil viscosity and promotes separation of the oil and water.

(9) In some embodiments, a heater treater includes a coalescing section that can include a spreader and an electrostatic device that passes alternating current through the emulsion to promote formation of water droplets, which promotes separation of the water droplets by gravity. The remaining “dry” oil can be removed from the heater treater by an oil outlet at an appropriate location on the heater treater unit.

(10) Many heater treater designs have been developed, including vertical and horizontal configurations, the choice of which depends on the particular desired parameters, such as design throughput, cycle time, and like factors.

(11) Upon exiting the heater treater **1240**, the oil stream **1342** can go to a stabilizer tower **1250**. In general, a stabilizer tower, such as stabilizer tower **1250**, typically includes trays, structured packing, and/or random packing in a column to promote contact between the vapor and liquid phases, permitting the transfer of mass and heat from one phase to the other. The trays have orifices for dispersing the gas uniformly on the tray and through the liquid on the tray. Types of trays include valve, bubble cap, and perforated-types. Structured packing often includes perforated plates that are folded and/or welded together. Random packing is available in many sizes and geometric shapes.

(12) Partial fractionation or distillation of the oil often occurs in the stabilizer tower. The heavier components and higher hydrocarbons flow through the column as liquid. Some of the liquid from the bottom of the column is withdrawn and circulated through a reboiler in some configurations to add heat to the process. In the reboiler, the lighter components are driven off as a gas. At each tray or stage the rising gas performs a stripping operation such that the lighter components in the gas increase as the gas rises through the column. Pressure inside the stabilizer column can range typically between 0 to 20 PSIG (0 to 1.4 bars). Other configurations, such as a reflux system, additional heat exchangers, and like equipment and processing may be included.

(13) The stabilized oil stream **1352**, often comprising pentane and higher hydrocarbons (C5+), exits the base of stabilizer tower **1250**. Oil stream **1352** may then be stored in tank **1265** at or near atmospheric pressure for eventual transport to an oil refinery or like user.

(14) The term “swell” is often used to refer to the increase in volume of an in-ground reservoir fluid (that is, in-ground), which includes oil, when solvent molecules dissolve in the reservoir fluid. In this regard, reservoir oil swell can enhance recovery of oil trapped in inaccessible pore spaces. This specification uses the term “swell,” also referred to as “uplift,” to refer to the volumetric expansion of an oil stream flow rate during processing.

SUMMARY

(15) In various implementations, a system and method for conditioning live crude oil decreases fugitive emissions, or increases volumetric oil output, or both decreases fugitive emissions and increases volumetric oil output relative to prior art systems. A system for conditioning live crude oil can include a separator, a stabilizer tower, and a heater treater that includes feeding a heater treater output gas to the stabilizer tower.

(16) The separator is adapted for receiving live crude oil from a wellhead and for producing a separator oil output and a separator gas output. The separator in some cases is considered part of the wellhead production facility. The stabilizer tower is adapted for (i) receiving the separator oil output and receiving a heater treater gas output and (ii) producing a stabilizer tower oil output and a stabilizer tower gas output.

(17) The heater treater is adapted for (i) receiving the stabilizer tower oil output and (ii) producing a heater treater oil output and a heater treater gas output; wherein the heater treater gas output has a temperature that is higher than the stabilizer tower gas output and the heater treater oil output is

stabilized oil. A portion of the heater treater oil output may be recycled to a heater treater inlet. The heater treater may also produce a heater treater water output. A portion of the heater treater water output may be commingled with a portion of the heater treater oil output to be recycled to a heater treater inlet.

(18) The system for conditioning live crude oil can include a vapor recovery unit (VRU) adapted for (i) receiving the stabilizer tower gas output and (ii) producing a VRU gas output and a VRU oil output. The heater treater is adapted for receiving the VRU oil output. As an alternative, the stabilizer tower is adapted for receiving the VRU oil output. The VRU can include discrete components and/or a packaged compressor and accessory components, while still being a VRU as used herein.

(19) The heater treater may be adapted for recirculating a recirculating portion of the heater treater oil output and/or the commingled heater treater oil output and heater treater water output into the heater treater. And the recirculating portion of the heater treater oil output may be combined with the VRU oil output upon or before entering the heater treater. The system may yield an oil volumetric production rate output (that is, stabilized oil that may be measured at a stabilized oil tank) that is greater than a volumetric production rate of the separator oil output, wherein the volumetric production rates are measured in BOPD.

(20) The live crude oil fed to the conditioning system includes at least an oil component and a gas component, and typically also includes a water component. Thus, the components disclosed herein may be two phase or three phase components. Typically, some aspect of the system will include a water separation capability.

(21) Hydrocarbon gas from the separator and/or from the stabilizer may be sent to at least one of a user, the stabilizer tower, and the heater treater. The stabilizer components may be pre-assembled (that is, in a fabrication facility) and mounted on a skid (that is, a unitary structural steel frame). The heater treater components may also be pre-assembled and mounted on a skid.

(22) The process for conditioning oil, often including increasing swell or uplift, can include steps for operating the system as described (in whole or in part) herein, including providing gas from the heater treater directly to the separator. The process for conditioning live crude oil may include the steps of: receiving a live crude oil stream from a wellhead into a separator, the live crude oil stream including at least an oil component and a gas component; separating a first gas stream from the live crude oil in the separator to create at least a first oil stream; receiving the first oil stream from the separator into a stabilizer tower; separating a second gas stream from the first oil stream in the stabilizer tower to create a second oil stream; receiving the second oil stream into a heater treater; separating a third gas stream from the second oil stream in the heater treater to create a stabilized oil stream and a third oil stream, and circulating the third gas stream from the heater treater to the stabilizer wherein the third gas stream combines with the second gas stream to create a combined second gas stream, the combined second gas stream flowing to a vapor recovery compressor; moving the stabilized oil stream to a stabilized oil tank; and circulating the third oil stream within the heater treater. The stabilized oil stream has a greater volumetric flow rate, measured in BBLD, than the volumetric flow rate of the stabilizer oil input, measured in BBLD.

(23) The process may include the step of circulating a recirculating portion of the second gas stream from the stabilizer tower to the heater treater and a conditioned portion of the second gas stream from the stabilizer tower to a user, and may include a step of circulating a recirculating portion of the VRU output gas stream to either the heater treater or stabilizer tower.

(24) The process may include the step of flowing the stabilized oil stream to a stabilized crude oil tank that is approximately at atmospheric pressure, or the process may include the step of flowing the stabilized oil stream directly to an end user. The step of circulating the third oil stream includes inputting the third oil stream at an inlet of the heater treater.

(25) The process may include the step of cooling the stabilized oil stream before it enters the stabilized crude oil tank. In some implementations, the step of cooling the stabilized oil stream may

be used to decrease emissions.

(26) The process for conditioning oil to decrease emissions, without increasing swell or uplift, does not require the step of combining the recirculating portion of the heater treater oil output with the VRU oil output upon or before entering the heater treater.

(27) The word stream does not require that the process be perfectly continuous or steady state. For merely one example, dump valves may operate in the equipment such that they close temporarily in response to liquid level in a unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A (Prior Art) is a process flow diagram of a conventional live crude oil stabilization process.

(2) FIG. 1B (Prior Art) shows a simple diagram of a traditional production facility using a vapor recovery tower to capture emissions off produced oil. This process describes a traditional upstream production facility process to eliminate emissions. From the wellhead, crude oil stream **1320** flows into a 3-phase separator **1230**. Produced oil stream **1332** flows into a 3-phase heater treater **1346**. Using applied heat and retention time, the inlet fluids to the heater treater will produce oil stream **1356**, water stream **1330**, and a gas stream **1334**. Oil stream **1356** flows into a vapor recovery tower **1746**. A vapory recovery tower is typically a 2-phase vertical tower, separating gas from oil. A vapor recovery tower can sometimes produce water. By operating the vapor recovery tower at a lower pressure than the upstream heater treater, this allows gas to separate from the inlet oil fluid. A vapor recovery tower is typically an open vessel with an internal pipe upcomer to drain oil from the bottom of the vessel into an oil storage tank. Sometimes there is a mesh pad in the tower.

(3) Produced oil stream **1456** will then flow into the oil storage tank **1646**. Gas stream **1458** produced from the vapor recovery tower **1746** will typically be comingled with the produced gas **1334** from heater treater **1346** and will be sent to the vapor recovery compressor **1430**. The vapor recovery compressor **1430** will typically control the pressure of the vapor recovery tower. The produced gas **1334** from heater treater **1346** can sometimes go straight to sales gas instead of to vapor recovery compressor **1430**. Vapor recovery compressor **1430** will compress the inlet gas and add line pressure so that it may be comingled with the higher pressure gas produced from 3-phase separator **1230** and be sent to gas sales.

(4) FIG. 1C shows a simple diagram of a production facility using an oil stabilization tower to capture emissions of produced oil. This figure is similar to FIG. 1B except it uses a stabilizer tower **1446** to release gas from produced oil stream **1332** by operating at low pressure.

(5) FIG. 1D shows a simple diagram of a production facility using an oil stabilization tower to capture emissions of produced oil. This figure is similar to FIG. 1C except the stabilizer tower **1446** is downstream of the heater treater **1346** and the pressure is not equalized between the two vessels. Produced oil from 3-phase separator **1230** sends oil to heater treater in stream **1332**. Heater treater will operate at a pressure to send the produced oil stream **1356** to the stabilizer tower. Produced gas **1334** from the heater treater **1346** will flow to the bottom of the stabilizer tower similar to FIG. 1C. Produced oil **1456** from stabilizer tower **1446** will flow directly to oil cooling. Gas stream **1434** from the vapor recovery compressor is sent to a gas cooling process to extract liquids of the gas stream.

(6) FIG. 1E shows a simple diagram of a production facility using a heater treater to capture emissions of produced oil. This figure is similar to FIG. 1D except the heater treater **1346** will act as the low pressure vessel and a stabilizer tower is not needed. Heater treater **1346** will operate at low pressure and will allow gas to separate from the oil. Produced gas in stream **1334** will go through the same liquid recovery process, compressing and cooling the gas to collect condensate.

Condensate **1634** will be sent from 2-phase separator **1520** back to the heater treater. The condensate will vaporize and mix with the low pressure vapor space inside heater treater **1346** and cause heavy hydrocarbons to fall out of the gas and add swell and an uplift in oil production. Produced oil stream **1356** from heater treater **1346** will be sent to oil cooling to stabilize the oil stream before entering oil storage tank **1646**.

(7) FIG. **1F** shows a simple diagram of a production facility using a vapor recovery tower to capture emissions of produced oil. This figure is similar to FIG. **1D**. Produced oil from 3-phase separator **1230** sends oil to heater treater in stream **1332**. Heater treater will operate at a pressure to send the produced oil stream **1356** to the vapor recovery tower **1746**. Produced gas from the heater treater **1346** will flow directly to the vapor recovery compressor **1430** to be compressed and cooled. Condensate **1634** produced from the 2-phase separator **1520** will be sent to the heater treater **1346** to cross exchange with the heater treater vapor space. Produced oil stream **1456** from vapor recovery tower **1746** will flow directly to oil cooling. Gas stream **1458** produced from vapor recovery tower **1746** will be comingled with gas stream **1334** from the heater treater to be sent directly to the vapor recovery compressor.

(8) FIG. **2A** is a simplified flow process diagram of a first portion of a first example of a live crude oil conditioning process using a 3-phase separator.

(9) FIG. **2B** is a simplified flow process diagram of a second portion, down-stream of the portion shown in FIG. **2A**, of the example of a live crude oil conditioning process.

(10) FIG. **2C** shows an example for another type of a production facility. This involves a 3-phase separator that sends separator oil from the wellhead emulsion to a 3-phase heater treater. A heater treater is used as secondary treatment for the produced oil. A heater treater vessel adds heat to the fluids and gas inside and to get further separation of water and gas out of the oil. Gas from the heater treater typically comingles with the cold 3-phase separator and goes to gas sales or goes to a vapor recovery compressor to raise pipeline pressure to send gas to sales if the heater treater is not operating at a sufficient pressure to send to sales.

(11) FIG. **2D** shows a basic flow diagram of the zero emissions system of the present disclosure, which includes a stabilizer tower upstream of a heater treater. Oil and water pumps recirculate fluids in the heater treater and transfer oil to oil cooling and oil storage, and water to water storage. A first make up gas stream **164b** is used to balance the stabilizer tower and add gas to the tower and keep a desired oil to gas ratio inside the tower. This keeps an efficient amount of gas inside the tower to contact the oil stream when oil inlet volumes decrease. Oil stream **153** is a stabilized oil stream from oil cooling to oil storage. Water pump **62** transfers water from heater treater to either water storage or will recirculate water to the inlet of the heater treater.

(12) FIG. **2E** shows a basic flow diagram of the zero emissions system of the present disclosure and is similar to FIG. **1D**. Heater treater upstream of the oil stabilizer tower. The heater treater internal pressure will send gas and oil to the stabilizer tower. Oil pumps recirculate fluids in the tower and transfer oil to oil cooling and oil storage. The oil stabilizer is the only vessel operating at low pressure.

(13) FIG. **3A** is a flow process diagram of a second example of a live crude conditioning process.

(14) FIG. **3B** is an enlarged portion of the flow process diagram of FIG. **3A**.

(15) FIG. **3C** shows a typical production facility process flow diagram with an oil stabilization tower process to reach zero tank emissions. Oil stream **322b** bypasses stabilizer tower to send oil to heater treater inlet. Water pump **251** transfers water to water storage or recirculates water to heater treater inlet. Oil pump **253** transfers oil to oil storage or recirculates oil to heater treater inlet. Gas stream **362** from a compressor unit that injects high pressure gas down the tubing of a well head for artificial wellhead lift in attempt to add pressure in a well to get fluids to rise up casing of well head and transfer the emulsion produced from a wellhead to the separator. Gas stream **363** sent to a gas cooling process from the vapor recovery unit **260** to extract liquids out of the gas stream.

(16) FIG. **3D** is similar to the facility process described in FIG. **3C** except using the process

described in FIG. 1D and FIG. 2E.

(17) FIG. 3E is similar to the facility process described in FIG. 3C except using the process described in FIG. 1E.

(18) FIG. 3F is similar to the facility process described in FIG. 3C except using the process described in FIG. 1F.

(19) FIG. 4A shows a process for reducing emissions. This process produces reduced emissions and minimizes swell. Since swell is not generated from the compression and cooling process off the vapor recovery compressor, oil cooling is not needed to reduce emissions. Oil outlet stream **352** sends oil to oil storage from heater treater. Recirculating water stream **356b** from water pump on heater treater.

(20) FIG. 4B shows a process for reducing emissions. Oil cooling is not always needed as the heater treater **250** temperature can be increased to make more gas escape from the oil phase in the process and reach near to at zero emissions.

(21) FIG. 4C shows a process for reducing emissions. This process is used and described in FIG. 1C, FIG. 2D, FIG. 3C.

(22) FIG. 4D shows a process for reducing emissions. This process is used and described in FIG. 1D, FIG. 2E, FIG. 3D.

(23) FIG. 4E shows a process for reducing emissions. This process is used and described in FIG. 1E and FIG. 3E.

(24) FIG. 4F shows a process for reducing emissions. This process is used and described in FIG. 1F and FIG. 3F.

(25) FIG. 5A shows a process for reducing emissions. This process includes using existing equipment, such as a vapor recovery tower **1746**. In this process, the vapor recovery tower oil outlet stream **342** may be routed directly to oil sales **353** using a LACT unit **271**. This process allows the vapor recovery oil outlet stream **342** to bypass the oil storage tank **270**, thereby eliminating the chance of fugitive emissions that are typically produced in oil storage tanks. In some implementations, operating the vapor recovery tower **240** at low pressure and bypassing the oil storage tank **270** can decrease emissions. In some implementations, oil produced by the VRU and gas cooling process may be sent to the heater treater inlet. In more detail, gas stream **363** flows through gas cooler **266** and separator **268**. The liquid that is collected by separator **268** then flows to heater treater **250**.

(26) FIG. 5B shows a process for reducing emissions. This process is similar to the process shown in FIG. 5A, except it uses an oil stabilizer tower **240** instead of a vapor recovery tower **1746** and further includes a two-phase oil storage vessel **275** through which the oil stabilizer tower oil outlet stream **342** is routed. The LACT unit **271** receives the oil outlet stream from the oil storage vessel **275** and sends it directly to oil sales **353**. Like the process shown in FIG. 5A, the oil storage tank **270** is bypassed to eliminate fugitive emissions that are typically produced in oil storage tanks.

(27) FIG. 6 (Prior Art) shows a simple diagram of a production facility. This figure is similar to FIG. 1B except it shows a stream of oil **1656** can be drawn from the stabilized crude oil storage tank **1646** and sent to an end user.

(28) FIG. 6A shows a process for reducing emissions. This figure is similar to FIG. 1D, FIG. 2D, FIG. 3B and FIG. 4D except the system may rely on gravity to flow third oil stream **152a** to the inlet of the stabilized crude oil storage tank **70**. In addition, oil cooling is removed from this process. A fourth oil stream **163** may transferred with the use of pressure from the VRU **60** to the heater treater **50**. Oil recirculating and transfer pump **52** may optionally be included to transfer recirculating oil stream **152b** to the inlet of stabilizer tower **40** and/or to transfer third oil stream **152a** to storage tank **70**.

(29) FIG. 6B shows a process for reducing emissions. This figure is similar to FIG. 6A but shows how the system would operate using control valves and with different operating pressures and temperatures throughout the process.

(30) FIG. 6C shows a process for reducing emissions. This figure is similar to FIG. 6A but includes a second stabilizer tower **41** that may receive first water stream **138** from the separator **30** and/or second water stream **148** from the heater **50** to produce fifth gas stream **184** that routes to the VRU **60**, fifth oil stream **182** that gravity flows to the stabilized crude oil storage tank **70**, and third water stream **183** that gravity flows to a stabilized water storage tank **71**.

(31) FIG. 6D shows an isometric view of a stabilizer tower **40**, **41** of the present disclosure, which may also be referred to herein as a vapor and liquid recovery tower or VLRT, as well as inlet lines for streams flowing into the VLRT and outlet lines for streams flowing out of the VLRT. The VLRT of the present disclosure may be used as the stabilizer towers **40**, **41** in the processes of FIG. 6A through FIG. 6C, for example.

(32) FIG. 6E shows a side cross-sectional view of an upper portion of the VLRT, which includes a top packing section **430** and a bottom packing section **435**.

(33) FIG. 6F shows an isometric view of a representative partial system **1000** that includes the VLRT that receives the second oil stream **142a** from the heater treater **50** and the second gas stream **154a** from the heater treater **50**, and that produces third oil stream **152a**, which that can be routed directly to one or more oil storage tanks **70**, or can be routed to a LACT unit **60**.

(34) FIG. 6G shows an isometric view of the representative partial system **1000** of FIG. 6F from a different vantage point, rotated approximately 180 degrees from the view of the system **1000** shown in FIG. 6F.

DETAILED DESCRIPTION

(35) Stabilizer Tower to Heater to Tank System

(36) To illustrate a first example of a system for stabilizing crude oil, a system **10** for stabilizing live crude oil includes a separator **30**, a stabilizer such as a stabilizer tower **40**, a heater treater **50**, a vapor recovery unit **60**, a stabilized oil tank **70**, and an oil and gas recirculation system **80**.

(37) As illustrated in FIG. 2A, separator **30** receives a first oil stream **122** from a wellhead **20** at inlet **32**. First oil stream or live crude feed **122** typically includes an emulsion of oil, gas, and water directly from wellhead **20**. The terms “first oil stream” and “live crude oil” encompasses any conventional wellhead pressures and temperatures and composition of hydrocarbons, according to API specifications. In this regard, wellhead pressures range from 2,000 to 150,000 psi (138 to 10,300 bars). Further the terms “first oil stream” and “live crude oil” encompass any conventional oil and gas stream from a wellhead, including but not limited to emulsions, such as with water or other liquid. It is understood that several wellheads **20** can feed a single separator **30**, and the symbol in FIG. 2A for separator **30** can represent several separators in parallel.

(38) Separator **30** is illustrated in FIG. 2A as a vertical separator, including a stream carrying live crude **122** to the top half of the separator **30**. The separator **30** produces a separator oil outlet stream **132** (also referred to herein as a first oil stream) at an oil stream outlet **33**, a separator gas outlet stream **134** (also referred to herein as a first gas stream) at a gas stream outlet **35**, and optionally a separator water outlet stream **136** (also referred to herein as a first water stream) at a water outlet **37**. The separator water outlet stream is optional, as system **10** encompasses two phase and three phase separators.

(39) Separator **30** may, in some vertical, three-phase configurations, include an inlet diverter and a mist eliminator, an oil level controller and oil dump valve, and a water dump valve. Separator **30** may also (or alternatively) include a downcomer and spreader, an interface controller and water dump valve, and oil weir level controller and oil dump valve. Other configurations of separator **30** and/or multiple stages may be employed. Separator **30** is not limited to vertical separators, as other configurations, such as horizontal separators, may be employed. Separator **30** often is near the one or more wellheads **20**, often as close as can be conveniently located. Separator **30** often can be remotely located, such as a mile from the wellhead **20**.

(40) Stabilizer tower **40** yields a stabilizer oil output stream **142**, also referred to as second oil output stream **142**, from an oil outlet **43**. Accordingly, stabilizer tower **40** can include a liquid level

controller and corresponding valves and instrumentation for operating stabilizer tower **40** as a two-phase process.

(41) The design features of separator **30** may be chosen and designed according to the process conditions, such as pressure, temperature, and live crude feed characteristics, and according to industry standards, as will be understood by persons familiar with oil and gas stabilization. Further, it is understood that separator **30** may include piping, valves, controls, and the like to perform its separation function, such as a gas back pressure valve, flare valve, a gas flow measurement device, and the like in the separator outlet gas stream piping.

(42) Of the separator output streams, gas stream **134** is typically suitable for use and can thus be sold to end users, and water stream **136** typically goes for water treatment, reinjection, or the like. As illustrated in FIG. 2B, oil stream **132** goes to stabilizer tower **40**.

(43) Pressure within stabilizer tower **40** typically is controlled by a gas back-pressure valve (or the like) to a pressure that often is no more than approximately 200 psi (14 bar). The liquid within stabilizer tower **40** flows by gravity through a series of trays, packing, and/or other media for stripping of gas from the liquid. In this regard, the internal components of stabilizer tower **40** may be chosen and configured in any way, as will be understood by persons familiar with oil stabilization and stabilizer tower technology.

(44) As described more fully below, stabilizer tower **40** includes an inlet **82** for receiving a heater treater gas output stream **154**. Thus, the gas output of stabilizer tower **40** is referred to as a combined gas stream **144**, also referred to as a combined second gas stream **144**, as a gas outlet **45**.

(45) Vapor recovery unit (VRU) **60** includes a compressor, often a screw type, that receives the combined gas stream **144** from stabilizer tower gas outlet **45**. VRU **60** can also include a demister, valves and controls, other conventional components. VRU packages are commercially available, as will be understood by persons familiar with oil stabilization technology.

(46) Liquid from the compression is discharged from VRU **60** at an oil outlet **63** to yield a VRU output oil stream **163** (that is, condensate), which can be controlled to be approximately at heater treater pressure. Oil stream **152b** enters into heater treater **50** at an oil inlet **52'**, which may be separate from heater treater inlet **52** that receives stabilizer tower oil output stream **142**.

(47) Gas that is pressurized to a desired pressure in VRU **60** is discharged at a gas outlet **61** to yield a VRU gas output stream **164a** that go be piped to an end user, accumulated with other gas streams, such as separator output gas stream **134**, and/or gas streams from other sources.

(48) Heater treater **50** is illustrated in FIG. 2B as a horizontal heater treater. Heater treater **50** includes an inlet **52** for receiving stabilizer oil output stream **142**, which typically is an emulsion of water, oil, and gas at approximately the stabilizer tower pressure. Heater treater **50** may include an oil dump valve, a gas back-pressure valve, a water dump valve and like process equipment and its instrumentation, as will be understood by persons familiar with heater treater technology in view of the information herein. Heater treater **50** may be of any type, such as vertical or horizontal, and may include combination of valves and their actuation, such as mechanical and pneumatic actuation. Chemical agents may be used to weaken the emulsifying agents, depending on the chemistry of the fluid in the heater treater, the process conditions, and the desired output properties.

(49) Heater treater **50** also includes a burner system **58** that typically includes a burner, a fire tube, a burner management system, and a stack. The burner management system includes a thermostat, a gas burner valve, and a safety system for controlling temperature in the process, such as fluid temperature within heater treater **50**. The fire tube is an indirect-type heat exchanger within heater treater **50** that transfers heat to the process fluid. The products of combustion exit the fire tube through the stack.

(50) Thus, after initial degassing in the inlet portion of heater treater **50** near inlet **52**, heat from the fire tube is transferred to the process fluid within heater treater **50**, which raises the process temperature to (typically) 100 to 160 degrees F. Heating the emulsion in this regard decreases fluid viscosity, enhances the separation of water from the oil, and promotes gas release. Gas from the

initial degassing and gas stripped from the emulsion via heating can be combined to yield a heater treater gas output stream **154**, which is also referred to herein as third gas stream **154**. As explained more fully below, gas output stream **154** is circulated back to recirculation gas inlet **82** of stabilizer tower **40** from a gas outlet **55** of heater treater **50**.

(51) Processing within heater treater **50** yields a stabilized oil output stream **152a** at an oil outlet **53** and a water output stream at water outlet **57**. Stabilized oil output stream **152a** is at a temperature and pressure that enables it to be sent to and stored in a stabilized crude oil tank **70** that is at atmospheric pressure.

(52) A portion, referred to herein as the oil recirculation stream **152b** and the third oil stream **152b**, of the oil output from heater treater **50** is recirculated from heater treater oil output **53** to oil inlet **52'** where preferably it is combined with VRU oil output stream **162**. As referred to above, the recirculation system **80** includes the oil recirculation stream **152b**. A pump **59** (shown in FIG. 2B) moves oil recirculation stream **152b** from heater treater oil outlet **53** to the second heater treater oil inlet **52'**. Recirculation of oil via oil recirculation stream **152b** is believed to enhance the conditioning process by increasing the volume of oil that is subject to treatment in heater treater **50**.

(53) Recirculation system **80** also includes gas recirculation stream **154** that is piped from heater treater gas outlet **55** to a stabilizer recirculation gas inlet **82**. Typically, heater treater pressure is greater than stabilizer tower pressure, such that gas recirculation stream **154** is moved via the pressure difference without requiring additional components, such as a compressor. Typical pressures in the stabilizer tower **40** and heater treater **50** typically are between 5 and 150 PSI (0.4 and 10.4 bars), according to the desired operating conditions.

(54) The inventors have demonstrated that oil stabilization system **10** and the associated process enhances the volumetric flow rate of stabilized oil stream **152a**. It is surmised that low pressure gas stream **154** from the heater treater flowing upwardly in stabilizer tower **40** in close contact with the oil emulsion dissolves or entrains gaseous hydrocarbons in the liquid stream, even while partial fractionation or distillation of the oil occurs in stabilizer tower **40** at typical stabilizer process conditions, such as 50 to 200 PSIG (3.4 to 14 bars), while retaining pentane and other higher hydrocarbons (such as C5+). Accordingly, it is believed that that fuel heating value and commercial value of stabilized oil stream **152a** is not unduly adversely affected.

(55) To illustrate a second example of system for conditioning crude oil, a system **210** for conditioning (stabilizing) live crude oil includes separators **230a**, **230b**, and **230c**, a stabilizer such as a stabilizer tower **240**, a heater treater **250**, a vapor recovery unit and scrubber **260**, a stabilized oil tank **270**, and an oil and gas recirculation system **280**. Each of the components of system **210**—including separators **230a-c**, stabilizer **240**, heater treater **250**, components of vapor recovery unit and **260**, and recirculation system **280**—have a structure and function as generally described with respect to corresponding components of first embodiment conditioning system **10**. System **210** further comprises sales gas scrubber **264**, artificial well gas lift compressor **262**, VRU discharge gas scrubber **268**, flare gas knockout **295** and **293**, water storage tank **273**, a high and medium pressure flare **291**, and tank vent gas combustor **292**.

(56) As illustrated in FIG. 3A, each of three wellheads **220a**, **220b**, and **220c** provide live crude to a corresponding separator **230a**, **230b**, and **230c** (respectively). In embodiment of FIG. 3A, as set out in Table 1, the total live oil feed **222** to the three separators **230a**, **230b**, and **230c** from the wellheads includes 3,450 BPD of oil and 6,000 MSCFD of gas. The live oil feeds in the embodiment of FIG. 3A has a pressure of 180 PSIG (12.4 bars) and a temperature of 90 degrees F. The outlets from **230a**, **230b**, and **230c** are illustrated as oil stream **322**, separator outlet gas stream **324**, and separator outlet water stream **326**. Oil stream **322** is at 90 degrees F. and has a pressure of 20 PSIG (1.4 bar), as the separator process results in a pressure decrease. Separators **230a**, **230b**, and **230c** in FIG. 3A preferably are conventional horizontal, three-phase separators.

(57) TABLE-US-00001 TABLE 1 STABILIZER TOWER TO HEATER TO TANKS TABLE 1 (IN REFERENCE TO FIG. 3A) STREAM/EQUIPMENT WELLHEAD OUTPUT 322 352a 326 369a

362 344 371 Item 1 2 3 4 5 6 7 8 INLET STAB STAB TOTAL GAS LIFT STAB OIL
TANK TOTAL OIL OIL WATER TO GAS TO TO 3 OVERHEAD FLASH FACILITY INLET
OUTLET TANKS USER WELLS GAS GAS BPD OIL 3,450 3,445 3,406 — — — — — BPD
WATER 6,900 3 — 6,897 — — — — — MSCFD 6,000 237 220 — 6,157 1,800 298 9 TEMP. F. 90
90 140 119 90 120 87 110 PRES. psig 180 20 6 17 175 1,200 5 1 STREAM/EQUIPMENT 359
364a&b 291 292 353 357 Item 9 10 11 12 13 14 RECYCLE LIQUIDS HP FLARE LP FLARE OIL
WATER OIL TO TO DESIGN DESIGN DELIVERY DELIVERY TO STAB STAB GAS GAS TO
LACT GATHERING BPD OIL 5,000 22 — — 3,260 — BPD WATER — — — — — 6,897
MSCFD — — 12,500 6,290 — — TEMP. F. 100 86 120 120 — — PRES. psig 20 25 100 1 — —

(58) Stabilizer tower **240** yields a stabilizer oil output stream **342** and a stabilizer gas outlet stream **340a** at 87 degrees F. and 5 PSIG (0.4 bar). As described more fully below, stabilizer tower **240** includes an inlet **282** for receiving a heater treater gas output stream **280a**. As illustrated in dashed line, heat treater gas output stream **354'** may provide a bypass or a partial bypass around stabilizer **240** for all or a portion of gas stream **354**. Gas stream **354'** or stabilizer tower **240** output gas stream **344** may bypass VRU **260** by flowing all or a portion of gas streams **354'** and **344** to flare gas knockout **295**. Flare gas knockout **295** yields a condensate output stream **396** controlled by liquid pump **296** and a gas output stream **395** to flare **291**. Condensate stream **396** flows to stabilized oil tank **270**. Oil storage tank **270** and water storage tank **273** yield a gas output stream **371** and **372** respectively. Gas streams **371** and **372** flow to flare gas knockout **293**. Flare knockout **293** produces a condensate stream **394** that is controlled by liquid pump **294**, and combines with condensate stream **396** to flow to oil storage tank **270**, and a gas stream **393** that flows to combustor **292**. Combustor **292** and flare **291** may be a single flare or combustor or a combination of both or like devices. Water storage tanks produce a water output stream **357** that is controlled by pump **274** to flow water stream **357** to a user.

(59) Vapor recovery unit (VRU) **260** includes a pair of packaged vapor recovery units and a vapor recovery scrubber **10**. Condensate stream **364a** from a gas lift compressor **262** (FIG. 3A) and other process equipment, such as condensate **364b** from 2-phase separator **264**, are fed into stabilizer tower **240**. Condensate streams **364a** and **364b** in the embodiment shown is 22 BPD at 86 degrees F. and 25 PSIG (1.7 bar). Sales gas scrubber **264** yields a gas outlet stream **369a** that can go to an end user for further processing or may produce a gas stream **369b** to gas flare **291**. A portion of gas stream **369a** is sent to gas lift compressor **262** to supply gas stream **362** to wellheads **220a**, **220b**, and **220c** for artificial well lift. VRU **260** yields an outlet stream **363** that is comprised of oil and gases that will feed into a two phase separator **268**. Separator **268** yields an oil output stream **362** that recirculate back to heater treater **250**. Separator **268** yields a gas output stream **367** that combines with gas stream **324**.

(60) Heater treater **250** receives stabilizer oil output stream **342**. Heater treater **250** yields a gas output stream **354**, which as explained above preferably is inserted into stabilizer tower **240** to form recirculation system **280**. Heater treater **250** also yields a heater treater oil output stream **352a** via an oil pump **253** and a heater treater water output stream **356** via water pump **257**. Heater treater oil output stream **352a** (that is, the stabilized oil output of the system **210**) is 3,406 BPD at 140 degrees F. and 6 PSIG (0.41 bar). Stabilized oil output stream **352a** is moved by oil pump **253** to stabilized oil tank **270**. The rate of oil stream **353** from tank **370** (item **13** in Table 1 and FIG. 3A) is a factor of the capability of the Lease Automatic Custody Transfer Unit (LACT) and/or downstream customer limitation.

(61) A portion of the heater treater output, an oil recirculation stream **352b** may be recirculated from a heater treater oil output to oil inlet of the heater treater **250**, as controlled by oil pump **253**. A portion of the heater treater water output, a water recirculation stream **353**, may also be recirculated from the heater treater **250** water output stream **356**, as controlled by water pump **257**.

(62) An optional recirculation system **358**, including an oil pump **259**, may circulate stabilized oil from tank **270** to stabilizer **240**, as needed to enhance the temperature, pressure, and/or other

variables relating to the system. In the embodiment of FIG. 3A, oil recirculation stream 359 is optional and can yield approximately 5,000 GPD at 100 degrees F. and 20 PSIG (1.4 bar). Oil tank output 353 in the embodiment shown is 3,260 BPD.

(63) In some embodiments, compressed gas stream 1434 from the vapor recovery compressor 1430 will be sent to a gas cooling process 1420 (See e.g., FIG. 1C, FIG. 1D, FIG. 1E, and FIG. 1f). Gas cooling can be done typically by a fan and belt cooler and radiator, or sometimes a heat exchanger that will cross exchange hot gas stream 1434 with a cold fluid or gas in attempt to lower gas stream 1434 temperature. Once the desired temperature is reached in gas stream 1434, it will flow into 2-phase separator 1520. By cooling the compressed gas to a desired temperature, it will allow heavy hydrocarbons such as pentanes and other C5+ hydrocarbons to compress into a liquid phase. 2-phase separator 1520 will separate gas from this hydrocarbon rich condensate. The produced gas from 2-phase separator 1520 will be sent to gas sales and the condensate 1634 captured will be returned back to stabilizer tower 1446. Condensate 1634 will enter the bottom of stabilizer tower 1446. When the condensate enters the tower, it will vaporize and rise up the tower to cross exchange with the oil stream coming down the tower in the packaging sections. This will allow the vapor pressure on the hydrocarbon condensate vapor to change and allow the heavy C5+ hydrocarbons to convert into a liquid phase. This process will cause an increased volumetric flow of hydrocarbon liquids in the stabilizer tower and will increase the swell created from the stabilizer tower. Some vapor will remain in the gas phase and comeingle with gas stream 1334 to be sent to vapor recovery compressor 1430. This will allow a loop of gas to be recycled to the inlet of the vapor recovery compressor and allow the compressor to have more of a constant flow of gas and keep the compressor running instead of shutting down due to lack of inlet pressure and flow.

(64) In some embodiments, the produced oil 1456 from heater treater 1346 will be sent to an oil cooling process 1546 (See e.g., FIG. 1C, FIG. 1D, FIG. 1E, and FIG. 1F). Oil cooling can be done with a fan and belt radiator or a heat exchanger that will cross exchange the hot oil will either a cold gas or liquid to cool oil stream 1456 to a desired temperature. After cooling to a desired temperature, oil stream 1556 will flow into an oil storage tank. Cooling the oil from heater treater 1346 will stabilize the crude oil and stop any gas from separating from oil stream 1556 when it enters a low pressure oil storage tank.

(65) The inventors have demonstrated that oil stabilization process 10 enhances the volumetric flow rate of stabilized oil stream 152b. It is surmised that low pressure gas stream 154 from the heater treater flowing upwardly in stabilizer tower 40 in close contact with the oil emulsion dissolves or entrains gaseous hydrocarbons in the liquid stream, even while partial fractionation or distillation of the oil occurs in stabilizer tower 40 at typical stabilizer process conditions (temperature and pressure) while retaining pentane and other higher hydrocarbons (such as C5+). Accordingly, it is believed that that fuel heating value of stabilized oil stream 152a is not unduly adversely affected.

(66) In this regard, the following process flow data has been calculated, based on a typical live crude oil stream 122, to compare a prior art stabilization system to the stabilization method of system 10.

(67) TABLE-US-00002 TABLE 2 STABILIZER TOWER TO HEATER TO TANKS Wellhead

Prior Art System	10	Output	Output	Output	Change	Oil	6703	6,566	6,684	+1.8%	BOPD	Oil
Output	0	-137	-19	118	Loss	BOPD	RVP	10	8	25%	PSIG	Gas
MMscfd	Water	14,510	14,510	14,510	—	BWPD						

(68) The prior art stabilization system in the second data column above is based on a conventional stabilizer model employing a first stage separator operating at 150 PSIG (10.3 bars), a heater treater operating at 50 PSIG and 120 degrees F., and a vapor recovery tower operating at 5 PSIG (0.4 bar). The data for stabilizer system 10 Output in the third data column above is based on a first stage separator 30 operating at 150 PSIG (10.3 bars), a stabilizer tower 40 operating at 6 PSIG, and a heater treater operating at 6 PSIG (0.41 bars) and 140 degrees F. The higher output temperature of

gas stream **154** from the heater treater **50** flowing into stabilizer **40** is believed to enhance the conditioning process.

(69) In this regard, the inventors understood that recirculation systems **80** and **280**, including gas streams **154** and **354** of system **10** and system **210**, enhances the stabilization process by (among other things) increasing the temperature in stabilizer tower **40** or **240** by introducing gas stream **154** or **354** from heater treater **50** or **250**. The inventors surmise that the increased temperature within tower **40** improves separation and retention of higher hydrocarbons (such as C5+) into the oil stream.

(70) The first row of Table 2 provides the oil output of the conventional stabilizer system and oil output of system **10** described herein—showing an improvement of in oil output per day of system **10** relative to the conventional stabilizer system. The second row of Table 20 provides the volumetric loss of oil from the available oil in the live crude from the first row. As shown, system **10** yields **118** more barrels per day more than the conventional stabilizer system, which is an improvement of approximately 1.8%. The units of Table 2 are million standard cubic feet of gas, barrels of oil per day, and barrels of water per day.

(71) The fourth row of Table 2 provides the gas output of the conventional stabilizer system and the gas output of system **10**—showing a decrease or “shrink” in gas production. In this regard, Table 2 reflects an increase in the volumetric flow rate of oil (that is, oil swell or uplift measured by stabilized oil stream **152a**) that is greater benefit than decrease in volumetric flow rate of the gas (that is, the sum of separator gas output stream **134** and VRU gas output stream **164**). Further, because of typical pricing structures in the oil and gas industry, a unit increase in stabilized oil production would outweigh a decrease in gas production of the same percentage magnitude. Thus, even if the magnitude of the percentage changes in were equal, system **10** would enhance the stabilization process compared with the conventional system.

(72) The third row of Table 2 provides the Reid Vapor Pressure (RVP) of the oil output. RVP is a property of the fuel at standard conditions—absolute vapor pressure exerted by the vapor of a liquid and any dissolved gases at 100 degrees F., according to test method ASTM-D323. Thus, RVP is a measure of the inherent volatility of the stabilized oil stream **152a** and correlates to losses of the gas output to the atmosphere. As reported in Table 2, RVP of the gas output from the conventional stabilizing system is reduced from 10 PSIG (0.7 bars to 8 PSIG (0.55 bars) by employing stabilizer system **10**.

(73) Fugitive emissions include leaks and other irregular releases of vapors or gasses from a pressurized processes, equipment, valves and piping, and the like. It is believed that the magnitude of fugitive emissions of hydrocarbons is related to pressure. Accordingly, the decrease in RVP, reflecting a decrease in actual pressure, of system **10** compared with that of the prior art (illustrated in Table 2) corresponds and illustrates a decrease in fugitive emissions of conditioning system **10**.

(74) Heater to Stabilizer Tower to End User

(75) To illustrate a third example, FIG. 5A depicts one implementation of a system **300** and FIG. 5B depicts another implementation of a system **310** for conditioning (stabilizing) live crude oil that enables an operator to selectively bypass the oil storage tanks **270** and instead route the stabilized oil directly to oil sales **353** using a LACT unit **271**.

(76) In more detail, FIG. 5A shows an implementation of a system **300** for reducing emissions. This system **300** includes using existing equipment, such as a 3-phase heated separator **250** and a vapor recovery tower **1746**. In this process, the inlet valve **276** to the oil storage tank **270** may be closed so that the vapor recovery tower oil outlet stream **342** may be routed directly to oil sales **353** using a LACT unit **271**. Because the vapor recovery oil outlet stream **342** bypasses the oil storage tank **270**, this process eliminates the chance of producing fugitive emissions within the oil storage tank **270**. In some implementations, the combination of operating the vapor recovery tower **1746** at low pressure and bypassing the oil storage tank **270** can further decrease emissions.

(77) FIG. 5B shows another implementation of a system **310** for reducing emissions. This system

310 is similar to the system **300** shown in FIG. 5A, except it uses an oil stabilizer tower **240** instead of a vapor recovery tower **1746** and further includes a two-phase oil storage vessel **275** through which the oil stabilizer tower oil outlet stream **342** may be routed to bypass the oil storage tank **270**. Thus, the LACT unit **271** can receive the oil outlet stream **344** from the oil storage vessel **275** and send it directly to oil sales **353**. Like the system **300** shown in FIG. 5A, the oil storage tank **270** is bypassed to eliminate fugitive emissions that are typically produced in oil storage tanks.

(78) FIG. 6 (Prior Art) shows a simple diagram of a production facility. This production facility is similar to the system shown in FIG. 1B, except FIG. 6 depicts that an oil outlet stream **1656** can be drawn from the stabilized crude oil storage tank **1646** and sent to an end user.

(79) Heater to Vapor and Liquid Recovery Tower to Tanks or End User

(80) To illustrate a fourth example, FIG. 6A depicts one implementation of a system **1000** and FIG. 6B depicts another implementation of a system **1010** for conditioning (stabilizing) live crude oil that enables an operator to selectively rely upon gravity to flow third oil stream **152a** to the inlet of the stabilized crude oil storage tank **70**.

(81) In more detail, FIG. 6A shows an implementation of a system **1000** that is similar to the systems shown in FIG. 1D, FIG. 2D, FIG. 3B and FIG. 4D except the stabilizer tower **40** used in system **1000** may be a vapor and liquid recovery tower (VLRT) as further described herein, and system **1000** may rely on gravity to flow third oil stream **152a** to the inlet of the stabilized crude oil storage tank **70**. In addition, oil cooling is removed from this system **1000**. A fourth oil stream **163** may be transferred with the use of pressure from the VRU **60** to the heater treater **50**. Oil recirculating and transfer pump **52** may optionally be included to transfer recirculating oil stream **152b** to the inlet of stabilizer tower **40** and/or to transfer third oil stream **152a** to the oil storage tank **70**.

(82) FIG. 6B shows another implementation of a system **1010** that is similar to the system shown in FIG. 6A but includes the use of control valves and with different operating pressures and temperatures throughout the process.

(83) FIG. 6C shows another implementation of a system **1020** that is similar to the systems shown in FIG. 6A except it includes a second stabilizer tower **41**, and the second stabilizer tower **41** used in system **1020** may be a vapor and liquid recovery tower (VLRT). The second VLRT **41** may receive first water stream **138** from the separator **30** and/or second water stream **148** from the heater **50** to produce fifth gas stream **184** that routes to the VRU **60**, fifth oil stream **182** that gravity flows to the stabilized crude oil storage tank **70**, and third water stream **183** that gravity flows to a stabilized water storage tank **71**.

(84) FIG. 6D shows an isometric view of a VLRT, as well as inlet lines for streams flowing into the VLRT and outlet lines for streams flowing out of the VLRT. When the VLRT is operating in the position of the first stabilizer tower **40** shown in FIG. 6A, FIG. 6B, and FIG. 6C, a first oil inlet line **402** to the VLRT will receive a produced oil stream, such as second oil stream **142a** from heater treater **50**, or alternatively, a produced oil stream from a crude oil/gas separation stage. The second oil stream **142a** will enter the top section of the VLRT and the oil will flow via gravity down through the VLRT. An oil outlet line **404** of the VLRT will receive processed third oil stream **152a** that may gravity feed without the use of a pump into one or more oil storage tanks **70** for holding. In one implementation, the oil outlet line **404** will be set to keep a minimum fluid level and will have a spill over outlet for third oil stream **152a** to flow to the top of an oil storage tank **70**. The third oil stream **152a** that enters oil storage tank **70** will be considered a stabilized oil stream such that the oil will have a reduced level of gas emissions while it is retained in the holding tank.

(85) A gas inlet line **406** to the VLRT will receive a produced gas stream, such as second gas stream **154a** from heater treater **50**, or alternatively, a produced gas stream from a crude oil/gas separation stage. The second gas stream **154a** will enter the VLRT in a lower section and rise up through the VLRT. A gas outlet line **408** of the VLRT will receive processed third gas stream **144a**

that flows to the inlet of the VRU **60** where the gas will be compressed and cooled. In some implementations, the gas will be cooled using an air-cooled fan cooler.

(86) As depicted in FIG. **6B**, the compression and cooling process in the VRU **60** will produce a combined liquid and gas stream **151** that will be processed through a 2-phase separator **68** to produce separate liquid and gas streams.

(87) The liquid stream produced from the separator **68** is referred to in FIG. **6A**, FIG. **6B** and FIG. **6C** as the fourth oil stream **163**. The liquid recovery process in the separator **68** will recover the light hydrocarbons, such as ethane, butane, and propane in the fourth oil stream **163**, which will be sent back to the heater treater **50**. Part of this fourth oil stream **163** will vaporize in the heater treater **50**, but will mix with the vapor stream inside the heater treater **50**. The vapor density of the condensate within the heater treater **50** will thereby get heavy enough to produce liquid droplets in the gas phase that drop out and mix with the liquid phase. The remaining vapors will proceed to the outlet of the heater treater **50** as the second gas stream **154a** to enter the gas inlet line **406** to the VLRT. The heater treater **50** provides a stable temperature of gas and liquid to be sent to the stabilizer tower.

(88) The gas stream **164a** from the separator **68** is routed to the end user, but a recirculating gas stream **164b** may be pulled from the gas stream **164a** for return to the heater treater **50** as shown in FIG. **6A**, FIG. **6B** and FIG. **6C**. Alternatively, the recirculating gas stream **164b** may be returned to the VLRT. This recirculating gas stream **164b** may be added to the VLRT to maintain a desired oil to gas ratio inside the packing sections of the VLRT, which are depicted in FIG. **6E** and described in more detail with respect to that drawing.

(89) In some implementations, an oil recirculation and transfer pump **52** may be installed on the bottom of the VLRT to recirculate oil stream **152b**, which is a portion of third oil stream **152a** output from the VLRT, back to the top of the VLRT through a second oil inlet **403**. The pump **52** may also transfer third oil stream **152a** to the oil storage tank **70**.

(90) The VLRT may include a spill over or flush out system to control a flooding situation in the top of the VLRT. If fluid starts to rise in the top section of the VLRT, an oil stream will be able to bypass the middle section of the VLRT and flow into the bottom section of the VLRT.

(91) In some implementations of the systems **1000**, **1010**, **1020**, second oil stream **142a** and/or second gas stream **154a** may optionally be routed to bypass the VLRT (stabilizer tower **40**) in case of an upset. In addition, the combined third gas stream **144a** may optionally be routed to bypass the VRU **60** in case the compressor is not operational or shuts down.

(92) FIG. **6E** shows a side cross-sectional view of an upper portion of the VLRT, which includes a top packing section **430** and a bottom packing section **435**. In some implementations, each packing section **430**, **435** is approximately 4 feet tall. Each packing section **430**, **435** may include a top coalescing baffle **431** and a bottom coalescing baffle **432**. These baffles **431**, **432** include holes so fluid can pass through, and in some implementations, the holes are approximately 0.5 inch in diameter. Between the top baffle **431** and bottom baffle **432** in each packing section **430**, **435** a packing material is employed to spread the oil stream out and increase contact surface area with the gas stream inside the VLRT. In some implementations, the packing material comprises 1 inch nominal diameter T-pac polypropylene random packing balls.

(93) As one of ordinary skill in the art will understand, the design of the upper portion of the VLRT may vary. In some implementations, packing trays may be used instead of coalescing baffles, pall rings may be used instead of random packing, the type of packing material may be altered, the height of the packing sections may be modified, the number of packing sections can vary, the dimensions of the baffle holes can change, the sizes of the packing material can be modified. Other design modifications are within the scope of the present disclosure.

(94) In operation, the second oil stream **142a** will flow into the VLRT through first oil inlet **402** and flow downwardly through the packing sections **430**, **435**. Likewise, the second gas stream **154a** will flow into the VLRT through the gas inlet **406** and rise upwardly through the packing sections

430, 435. In some implementations, when the second gas stream **154a** flows into the bottom section of the VLRT, a displacement nozzle **440** is employed to disperse the gas stream evenly in the VLRT before it flows through the packing sections **430, 435**. In some implementations, the displacement nozzle **440** comprises a pipe with holes drilled into the bottom of the pipe. The pipe may have a 4 inch nominal diameter and the drilled holes may be approximately 0.5 inches in diameter.

(95) Within the packing sections **430, 435**, the gas stream and the oil stream will cross-exchange such that the gas stream comes in contact with the oil stream. At a certain pressure and temperature ranges, heavy and light hydrocarbons (ethane through decane) molecules will be added to the oil stream such that the third oil stream **152a** exiting the VLRT has a larger volume than the second oil stream **142a** that enters the VLRT, and the combined third gas stream **144a** exiting the VLRT has a smaller volume than the second gas stream **154a** that enters the VLRT. In some implementations, the VLRT operates in a pressure range of approximately 0 psig to approximately 20 psig and a temperature range of approximately 100 degrees Fahrenheit to approximately 150 degrees Fahrenheit. The temperature and the pressure settings in the VLRT can be modified to alter the Reid Vapor Pressure (RVP) of the gas stream **144a** and the American Petroleum Institute (API) properties of the oil stream **152a** exiting the VLRT to meet the acceptance criteria of the end user.

(96) FIG. 6F shows an isometric view of a representative partial system **1000, 1010** that includes the VRU **60**, the VLRT, and a plurality of storage tanks **70**. FIG. 6G shows an isometric view of the representative partial system **1000, 1010, 1020** of FIG. 6F from a different vantage point, rotated approximately 180 degrees from the view of the system **1000, 1010, 1020** shown in FIG. 6F. As previously discussed, the VLRT receives the second oil stream **142a** from the heater treater **50** and the second gas stream **154a** from the heater treater **50** to produce third oil stream **152a**. As best shown in FIG. 6G, the third oil stream **152a** can be routed directly to one of the oil storage tanks **70**, or the third oil stream **152a** can become a bypass stream **152d** that bypasses the oil storage tanks **70** to be routed to a liquid transfer pump **71** shown in FIG. 6B. A plurality of lines may be coupled to the tops of the oil storage tanks to carry gas stream **179** to vent or to a flare to be burned and released.

(97) As previously described, the VLRT is designed to cross exchange an oil stream and a gas stream through a packing section that is configured to substantially maximize surface area contact between the oil stream and the gas stream. This process reduces fugitive emissions from the oil stream and recovers hydrocarbons that are typically carried with the gas stream, thereby producing a higher volume yield in the oil stream.

(98) The VLRT may also be adapted for receiving a water stream consisting of a water and oil and gas emulsion instead of only an oil and gas emulsion stream. In this configuration, the VLRT may produce a gas outlet stream, and both an oil outlet stream and a water outlet stream. This VLRT design can also be used to separate oil and gas out of a water inlet stream. In this case, the heater treater output gas stream will bypass the VLRT. Both the oil and water output stream will be gravity fed to an oil or water storage tank or to oil or water liquid transfer pumps. The VLRT may be used to extract gas from an oil stream or extract gas or gas and oil from a water stream.

(99) The VLRT design does not require the addition of energy to transfer the oil stream and gas stream through the equipment. Instead, the VLRT is operable to gravity feed the third oil stream **152a** either to a storage tank **70** or to a bypass stream **152d** that feeds the suction line of a liquid transfer pump **71**, all while reducing fugitive emissions and recovering hydrocarbons.

(100) In this regard, Table 3 below summarizes process flow data that has been calculated, based on a typical wellhead output, to compare the oil, gas, and water outputs of a prior art VRT to the oil, gas, and water outputs of the VLRT when the oil stream output is routed to an oil tank for storage before reaching an end user.

(101) TABLE-US-00003 TABLE 3 HEATER TO VAPOR AND LIQUID RECOVERY TOWER TO OIL TANK TO END USER Wellhead Prior Art New VLRT Output Output (VRT) Output

Change Oil 7,800 7,032 7,171 +1.97% BOPD Oil Output 0 -768 -599 +169 Loss BOPD RVP — 8.15 9.19 +1.04 PSIG API — 51.56 51.98 +0.42 Gas 11.00 12.40 12.19 -1.69% MMscfd Water 13,000 13,000 13,000 — BWPD

(102) Similarly, Table 4 below summarizes process flow data that has been calculated, based on a typical wellhead output, to compare the oil, gas, and water outputs of a prior art VRT to the oil, gas, and water outputs of the VLRT **400** when the oil stream output is routed directly to a liquid transfer pump, bypassing storage in oil tanks.

(103) TABLE-US-00004 TABLE 4 VAPOR AND LIQUID RECOVERY TOWER TO TRANSFER PUMP (NO OIL TANKS) Wellhead Prior Art New VLRT Output Output (VRT) Output Change Oil 7,800 7,032 7,201 +2.40% BOPD Oil Output 0 -768 -599 +169 Loss BOPD RVP — 8.15 9.51 +1.36 PSIG API — 51.56 52.16 +0.60 Gas 11.00 12.40 12.19 -1.69% MMscfd Water 13,000 13,000 13,000 — BWPD

(104) Table 5 below summarizes inlet and outlet conditions that have been calculated, based on a typical wellhead output with typical temperatures and pressures, for a system that includes the VLRT **400**. Table 5 shows how similar operating temperatures and pressures have different oil volumetric output for VLRT.

(105) TABLE-US-00005 TABLE 5 OPERATING CONDITIONS FOR VLRT INPUTS AND OUTPUTS Vapor and Liquid Recovery Tower Wellhead Emulsion 3-phase Heater (VLRT) Oil Gas Water separator Treater Temperature F. 100 100 100 97.28 140 Pressure psia 134.7 134.7 134.7 122.8 64.7 Liq Vol Flow @Std Cond barrel/day 7800 — 13000 — — Std Gas Flow MSCFD — 11,000 — — — Vapor and VLRT Oil Liquid Oil Tank Oil Tank Total Gas Outlet to Vapor and Liquid Recovery Tower Recovery Vent Gas Oil to End Transfer (VLRT) Tower Outlet Outlet User Pump Temperature F. 132.9 105 105 96.27 116 Pressure psia 15.7 14.67 14.67 115.9 15.7 Liq Vol Flow @Std Cond barrel/day — — 7170.64 — 7201.14 Std Gas Flow MSCFD — 1.4 — 12.19 —

(106) Table 6 below summarizes inlet and outlet conditions that have been calculated, based on a typical wellhead output with typical temperatures and pressures, for a system that includes a prior art Vapor Recovery Tower (VRT).

(107) TABLE-US-00006 TABLE 6 OPERATION CONDITIONS FOR VRT INPUTS AND OUTPUTS Vapor Recovery Tower Wellhead Emulsion 3-phase Heater (Prior Art) Oil Gas Water separator Treater Temperature F. 100 100 100 97.28 140 Pressure psia 134.7 134.7 134.7 122.8 64.7 Liq Vol Flow @Std Cond barrel/day 7800 — 13000 — — Std Gas Flow MSCFD — 11,000 — — — Oil Tank Oil Total Vapor Recovery Tower Vapor Recovery Vent Gas Tank Oil Gas to (Prior Art) Tower Outlet Outlet End User Temperature F. 122.4 105 105 96.27 Pressure psia 15.7 14.67 14.67 115.9 15.7 Liq Vol Flow @Std Cond barrel/day — — 7031.86 — Std Gas Flow MSCFD — 10.1 — 12.40

(108) A comparison between Table 5 and Table 6 shows how similar operating temperatures and pressures produce different oil volumetric output for VLRT.

(109) The systems and processes described herein refer to process flows from and to components, and/or that a component receives or is adapted to receive a process flow from another component. In this regard, these process flow terms encompass flow directly from the first specified component to the second specified component without major process equipment in between, but including piping, valves, pressure relief devices, safety and monitoring devices, instrumentation, and the like as needed. The description is not limited by prohibiting major process equipment or processes between the first specified component to the second specified component, as it is understood that components, sub-systems, and processes may be added between any of the components (such as wellhead **20** or **220a-c**, separator **30** or **230**, stabilizer tower **40** or **240**, heater treater **50** or **250**, VRU **60** or **260**, tank **70** or **270**, vapor recovery tower **1746**, VLRT **400**), and that the components can be modified in many ways, consistent with the broad conception of the invention and defined in the claims.

(110) The process data provided herein is design data; actual operating data may vary according to change in condition and/or desired output and the like, as will be understood by persons familiar with oil and gas processing technology. Further, the process data provided in the specification is or are examples which are not intended to limit the scope of the invention.

(111) The description herein describes particular examples of components, systems, and processes. The present invention is not limited to the particular components, systems, and processes specified herein. Rather, it is intended that the scope of the present invention be measured by the claims, without viewing any components, systems, or processes of the specification as essential. It is also understood that a person familiar with crude oil stabilization technology would understand that many terms used herein have established meaning that is specific to the oil and gas industry and/or oil stabilization technology, and that the terms inherently include many details that are not necessary to recite.

(112) Further, the information in the Background section describes conventional oil stabilization technology and components. It is not intended to disclaim any subject matter for any component, sub-system, or system, as the preferred embodiments described in the specification incorporate aspects of the conventional technology.

EXAMPLES

(113) The following examples are provided to further describe some of the implementations disclosed herein. These examples are intended to illustrate, not to limit, the disclosed implementations.

(114) FIG. 3C shows an exemplary 3-well production facility but is not limited to the number of wells used in a facility design. Each well head, **220a**, **200b**, **220c** will flow an emulsion fluid, a mixture of oil, gas, and water, to a 3-phase separator **230a**, **230b**, **230c** respectively, for the first stage of processing and separation. The 3-phase separator will separate the inlet emulsion into oil, gas, and water using gravity and retention time. Produced water from each 3-phase separator will comeingle and will flow into water storage tanks. Typically a gun barrel **272** is used to catch any oil that is carried over into the water stream. Water from gun barrel **272** is gravity fed into the water storage tank **273**. Oil produced in gun barrel **272** is gravity fed into the oil storage tank **270**. Water in the water tank **273** is pumped with pump **274** to a salt water disposal facility.

(115) Gas produced in 3-phase separator **230a**, **230b**, and **230c** will comeingle gas streams and flow in gas stream **334** to a 2-phase separator **264**. This 2-phase separator will capture any liquids and produced condensate from the gas stream. Condensate from 2-phase separator **264** will flow into the produced oil stream **322a** from the 3-phase separators. Gas leaving 2-phase separator **264** will be sent to sales to be purchased from the midstream purchaser in stream **369a**. If the midstream purchaser cannot accept the gas or if there is an issue sending gas down sales line, a back pressure valve holding pressure will open and send the gas down stream **369b** to be combusted using a flare.

(116) A desired amount of gas off stream **369a** will be sent to a gas compressor **262** that sometimes is used for artificial well head lift. This process involved sending high pressure gas down the tubing of a well head to add pressure downhole to supply force for fluids to rise up the well and enter the 3-phase separator. When gas on the compressor **262** is compressed, condensate is separated and from the gas stream and sent down condensate stream **364a** to comeingle with the oil stream **322a**.

(117) Oil produced from 3-phase separator **230a**, **230b**, and **230c** will be comeingled and sent to the maze process through oil stream **322a**. The maze process used in this example is similar to the process described in FIG. 1C and FIG. 2D. Oil stream **322a** will enter the stabilizer tower **240** that is equalized in pressure with heater treater **250**. There can be a bypass around the stabilizer tower for oil stream **322a** to enter directly to heater treater **250**. Oil flows down the stabilizer tower **240** and gravity feeds into heater treater **250**. As oil stabilizes in the stabilizer tower **240**, gas is separated from the oil and sent out the top of the tower in gas stream **340a** to the vapor recovery compressor **260**. Vapor recovery compressor **260** typically has an inlet 2-phase separator that will catch any carried over liquids in gas stream **340a**. This 2-phase separator will typically have a

pump of pressurized discharge process for liquids to be transferred. This liquid stream **368b** will be sent back to the oil stabilizer tower **240**.

(118) Gas stream **340a** will be compressed and sent to a gas cooling process **266** and then to a 2-phase separator **268** to collect condensation from the compression and cooling process. This condensation is sent from 2-phase separator **268** in stream **368a** to the stabilizer tower to cross exchange with the oil inside the tower. The gas produced from the 2-phase separator **268** will flow in stream **367a** to comingle with the produced gas stream **334** from the 3-phase separators to be sent to sales. There can be a back pressure valve on stream **367a** to control the discharge pressure of the vapor recovery compressor **260**. Stream **367b** will be a gas stream that will be sent to the stabilizer tower **240** to maintain a desired gas to oil ratio inside the tower. If the vapor recovery compressor **260** shuts off or cannot handle the flow rate, a back pressure valve will open and send the gas stream **340b** to a 2-phase separator **295**. This 2-phase separator will catch any liquids that are carried over in gas stream **340b** and send liquids to the oil storage tank **270** using a pump **296**. Gas from the 2-phase separator **295** will be sent to flare **291** to be combusted.

(119) Oil from the stabilizer tower **240** is gravity fed into heater treater **250** to be stabilized and process further and get an excess gas and water to be separated from the oil phase. Oil and water pumps are installed to operate a level control process on the heater treater **250**. Oil and water will recirculate in the heater treater until a desired height of fluids are reached. When this high water or oil level is reached, the level control process on heater treater **250** will transfer water stream **356** to the gun barrel **272** and oil to the oil cooling process to stabilize the oil further. Once oil stream **352a** is cooled, oil will be transferred to oil storage tank **270**. This oil in stream **352a** will be stabilized crude and will have zero emissions when it enters and is stored in oil storage tank **270**.

(120) Produced gas from the heater treater **250** will free flow into the bottom of the tower in gas stream **280a** and rise up the stabilizer tower **240** and cross exchange with the oil stream inside the tower. This allows contact point of oil and gas and allows the heavy hydrocarbons to retain in the liquid phase and add extra oil volume and swell to the process. If gas stream **280a** cannot enter the stabilizer tower **240**, a back pressure valve will open up and bypass the tower and send gas stream **280b** to the vapor recovery compressor **260** or the flare **291** if the vapor recovery compressor is shut off or over ran.

(121) Once oil is sent and stored in the oil storage tank, a Lease Allocation Custody Transfer unit monitors and controls the oil tank fluid level. Once oil fluid levels reach a desired level in oil storage tank **270**, the LACT unit measures and transfers oil down stream **353** to the midstream oil purchaser. All storage tanks including the gun barrel **272**, water storage tank **273** and oil storage tank **270** have vent lines comingled that transfer the produced gas to a combustor **292**. The combustor can also be a typical flare. A back pressure valve is installed on the vent line out to the flare to keep unnecessary flaring from tank in breathing. Tank in breathing is when storage tanks build up pressure solely from filling up with fluids and decreased with SWD pumps or LACT unit pumps drain the fluids from the storage tanks. To prevent flaring when storage tanks fill up with fluids, the back pressure will be set a desired pressure to accommodate the in breathing of tanks and prevent unnecessary gas from being sent to the flare. A flare knock out vessel **293** will capture any carry over fluids off the storage tank vent lines and send any fluids captured back to the oil storage tank **270** using pump **294**.

(122) Those skilled in the art will appreciate that numerous changes and modifications can be made to the preferred embodiments disclosed herein and that such changes and modifications can be made without departing from the spirit of the invention. It is, therefore, intended that the appended claims cover all such equivalent variations as fall within the true spirit and scope of the invention.

Claims

1. A system for conditioning a liquid stream comprising: a vapor and liquid recovery tower (VLRT) comprising: a top section adapted to receive a liquid stream with entrained gas that gravity flows down through the VLRT; and a packing section comprising packing material adapted to release the entrained gas from the liquid stream to produce a VLRT liquid output and a VLRT gas output; at least one of an oil storage tank and an oil liquid transfer pump; and at least one of a water storage tank and a water liquid transfer pump; wherein the liquid stream is a water emulsion, and the VLRT liquid output is a VLRT oil output and a VLRT water output; wherein the VLRT oil output gravity feeds out of the VLRT and is routed to the oil storage tank or to the oil liquid transfer pump; and wherein the VLRT water output gravity feeds out of the VLRT and is routed to the water storage tank or to the water liquid transfer pump.
2. The system of claim 1, wherein the VLRT further comprises: a lower section adapted to receive a gas stream that rises up through the packing section; wherein the packing material is further adapted to increase surface area contact between the liquid stream and the gas stream; and wherein the VLRT gas output comprises the entrained gas and the gas stream.
3. The system of claim 2, wherein the VLRT further comprises: a displacement nozzle adapted to disperse the gas stream received into the lower section before the gas stream rises up through the packing section.
4. The system of claim 1, wherein the packing material of the VLRT comprises random packing balls or pall rings.
5. The system of claim 4, wherein the VLRT further comprises: one or more baffles or packing trays adapted to contain the packing material.
6. A system for conditioning a liquid stream comprising: a vapor and liquid recovery tower (VLRT) comprising: a top section adapted to receive a liquid stream with entrained gas that gravity flows down through the VLRT; and a packing section comprising packing material adapted to release the entrained gas from the liquid stream to produce a VLRT liquid output and a VLRT gas output; a lower section adapted to receive a gas stream that rises up through the packing section; wherein the packing material is further adapted to increase surface area contact between the liquid stream and the gas stream; and wherein the VLRT gas output comprises the entrained gas and the gas stream; a separator adapted for receiving live crude oil from a wellhead and for producing a separator oil output and a separator gas output; and a heater treater adapted for (i) receiving the separator oil output and (ii) producing a heater treater oil output and the heater treater gas output; wherein the VLRT is adapted for (i) receiving the heater treater oil output as the liquid stream and receiving the heater treater gas output as the gas stream; and (ii) producing a VLRT oil output as the VLRT liquid output.
7. The system of claim 6, wherein the VLRT gas output has a lower volume than the heater treater gas output and the VLRT oil output has a higher volume than the heater treater oil output.
8. The system of claim 6, wherein: the VLRT oil output gravity feeds to an oil storage tank or a liquid transfer pump.
9. The system of claim 6, further comprising: a pump adapted for transferring a recirculating portion of the VLRT oil output back into the VLRT.
10. The system of claim 9, wherein the pump is further adapted for transferring the VLRT oil output to an oil storage tank or to a liquid transfer pump.
11. The system of claim 6, further comprising: a crude oil storage tank adapted to receive the VLRT oil output for stabilization.
12. The system of claim 6, further comprising: a liquid transfer pump adapted to (i) receive the VLRT oil output and (ii) provide the VLRT oil output to a user.
13. The system of claim 6, further comprising: a vapor recovery unit (VRU) adapted for (i) receiving the VLRT gas output and (ii) producing a VRU gas output and a VRU oil output.
14. The system of claim 13, wherein the heater treater is adapted for (i) receiving the VRU oil

output and (ii) receiving a recirculating portion of the VRU gas output.

15. The system of claim 13, wherein the separator is adapted for producing a separator gas output and the system is adapted for (i) combining the VRU gas output with the separator gas output and (ii) providing the combined gas output to a user.
